

Chapter 57: Gastrointestinal Disorders — Revision Notes

Williams Obstetrics 26e, pp. 2630–2674 of the PDF. High-yield numbers are in **bold**. The seven tables are transcribed from the PDF images (they are missing from the extracted chapter text). All 4 figures are included from the PDF.

Big picture: Pregnancy displaces and alters the GI tract, so symptoms, signs and the WBC become less reliable as gestation advances. Investigate and operate when indicated (ideally 2nd trimester for elective procedures): **delay in diagnosis, not the intervention, is what harms mother and fetus**, especially with appendicitis and bowel obstruction.

1. General Considerations

Diagnostic techniques

- **Endoscopy** (upper GI, ERCP, colonoscopy, sigmoidoscopy) can be used in pregnancy without radiation. Perform **when indicated, ideally in the 2nd trimester**, with a multidisciplinary team (obstetrics, GI, anesthesia).
 - Video capsule endoscopy: limited experience (slow GI transit; capsule electromagnetic field are theoretical concerns).
 - Colonoscopy indications: chronic abdominal pain, hematochezia, diarrhea.
 - **Bowel prep: polyethylene glycol electrolyte (GoLYTELY) or sodium sulfate (Suprep)** → avoids serious dehydration and ↓ uteroplacental perfusion. Tap water enemas are an alternative for rectosigmoidoscopy.
- **Noninvasive imaging:**
 - **Sonography is the ideal first technique**, but the gravid uterus obscures organs.
 - **MR imaging** (e.g., MRCP, MR enterography) avoids CT radiation.
- **Surgery:** intraabdominal surgery is the **most frequent nonobstetrical procedure** in pregnancy; **perforative appendicitis** is the most common lifesaving example. **Laparoscopy** has replaced open surgery for many conditions (ACOG 2019; SAGES guidelines).

Table 57-1. Preprocedural Considerations for Gastrointestinal Endoscopy During Pregnancy

Consideration
Plan consultation with an obstetrician, gastroenterologist, and anesthesiologist
Place patient in left lateral decubitus position
Use lowest effective dose of sedation necessary
Give prophylactic antibiotics as indicated. Penicillin, cephalosporin, erythromycin, and clindamycin are safe options
Minimize procedure time
Obtain fetal heart tones at the discretion of the obstetrician. In general, pre- and post-procedure heart tones are adequate
For colonoscopy, favor preparation with PEG-ES or with tap water enemas depending on GI level to be evaluated

GI = gastrointestinal; PEG-ES = polyethylene glycol electrolyte solution. From American Society for Gastrointestinal Endoscopy, 2012, 2015.

Nutritional support

- **Enteral (usually nasogastric) is preferred** — fewer serious complications. Very few obstetric conditions prohibit enteral feeding as a first effort.
 - Extreme cases (recalcitrant hyperemesis): **PEG with jejunal port (PEG-J)**.
- **Parenteral nutrition** when the gut must be quiescent:
 - **Peripheral (PPN)**: dilute solutions, short-term use.
 - **Central (CPN)**: hyperosmolar, needs central access; gives **24–40 kcal/kg/d**, mainly hypertonic glucose.
 - Most common CPN indications in pregnancy: **short bowel syndrome and dysmotility**; mean duration **33 days**.
- **Complications of parenteral nutrition**:
 - **Catheter sepsis is most common** (**25%** in 27 women with hyperemesis).
 - Upper-extremity venous thrombosis, volume overload, reversible liver dysfunction.
 - Fetal subdural hematoma from maternal **vitamin K deficiency** (rare).
- **PICC lines: 56% complication rate** (bacteremia most frequent) in 66 pregnant women. Reasonable for **short-term (few weeks)** nutrition.

Table 57-2. Some Conditions Treated with Enteral or Parenteral Nutrition During Pregnancy^a

Conditions		
Achalasia	Crohn disease	Ostomy obstruction
Anorexia nervosa	Diabetic gastropathy	Pancreatitis
Appendiceal rupture	Esophageal injury	Preeclampsia syndrome
Bowel obstruction	Hyperemesis gravidarum	Short bowel syndrome
Burns	Jejunioileal bypass	Stroke
Cholecystitis	Malignancies	Ulcerative colitis

^aThe printed footnote reads "Complications arranged alphabetically"; the entries are conditions, arranged alphabetically (read down each column). From Billiauws, 2017; Folk, 2004; Guglielmi, 2006; Mahadevan, 2019; Porter, 2014; Russo-Stieglitz, 1999; Saha, 2009; Spiliopoulos, 2013.

2. Hyperemesis Gravidarum

Definition and epidemiology

- Nausea and vomiting are common until **~16 wk**. Hyperemesis = **severe unrelenting nausea and vomiting** producing:
 - **Weight loss, dehydration, ketosis**
 - **Alkalosis** (loss of HCl) and **hypokalemia**
 - Acidosis from partial starvation
 - ± transient hepatic dysfunction and biliary sludge
- **A diagnosis of exclusion.**
- **Hospitalization rate 0.5–2.1%. Up to 25%** admitted in one pregnancy need admission again in the next.
- Etiology unknown, multifactorial: high or rapidly rising **hCG**, estrogen, progesterone, placental GH, prolactin, thyroxine, adrenocortical hormones; also ghrelin, leptin, nesfatin-1, peptide YY 3–36. Psychological components in some.

Risk factors

↑ Risk	↓ Risk
Hyperthyroidism, prior hyperemesis, diabetes, GI illness, restrictive diets, asthma	Smoking and obesity (↓ 1.5-fold)
Female fetus (perhaps estrogen-related)	
<i>H pylori</i> (proposed, inconclusive)	

- **Chronic marijuana use** → **cannabinoid hyperemesis syndrome** (mimic).
- Associations: anemia, **VTE**, gestational hypertension, preeclampsia. Link with preterm birth unclear.
- **No long-term maternal consequences** (no ↑ cardiovascular risk).

Complications

Table 57-3. Some Serious and Life-Threatening Complications Reported with Hyperemesis Gravidarum^a

Complication
Acute kidney injury —may require dialysis
Depression—cause versus effect?
Diaphragmatic rupture
Esophageal rupture—Boerhaave syndrome
Hyperalimentation complications
Hypokalemia —arrhythmias, cardiac arrest
Hypoprothrombinemia— vitamin K deficiency
Mallory–Weiss tears —bleeding, pneumothorax, pneumomediastinum, pneumopericardium
Rhabdomyolysis
Wernicke encephalopathy—thiamine deficiency

^aComplications arranged alphabetically.

- **Wernicke encephalopathy (thiamine deficiency):**
 - Triad **ocular signs + confusion + ataxia** in ~60%.
 - EEG may be abnormal; **MR imaging usually shows findings**.
 - Can recur in a later pregnancy. Maternal deaths; sequelae = **blindness, convulsions, coma**.
- **Vitamin K deficiency:** maternal coagulopathy, **fetal intracranial hemorrhage**, vitamin K embryopathy.
- AKI from dehydration (± rhabdomyolysis); rarely dialysis.

Management

- **First line (outpatient): doxylamine 10 mg + pyridoxine 10 mg (Diclegis)**
 - **2 tablets at bedtime** → add 1 first-morning tablet → then 1 morning + 1 midafternoon + 2 bedtime (max 4/d).
 - Parkland cost-saving version: **½ of a 25-mg Unisom (doxylamine) + 25-mg vitamin B6**; same graduated dosing, max 3 doses/d.
- **Ondansetron:** slightly more effective than doxylamine–pyridoxine. Not teratogenic on recent review, but risks **prolonged QT and serotonin syndrome** → **reserve for severe cases after 8 wk**.
- **IV crystalloid** for dehydration, ketonemia, electrolyte and acid–base deficits.
 - **Hypokalemia is common**; cardiac arrest reported.
 - **No benefit from adding 5% dextrose**.
 - **Thiamine 100 mg in 1 L crystalloid** for women needing IV hydration who have **vomited >3 weeks**; repeat daily for **2–3 days**, then IV multivitamins if IV fluids continue.

- **Hospitalize** if vomiting persists after rehydration and failed outpatient care: parenteral **promethazine, prochlorperazine, chlorpromazine or metoclopramide** (several prolong QT).
- **Glucocorticoids**: largely ineffective; **only for refractory cases** (putative teratogenicity).
- **Persistent vomiting** → **exclude other causes**: gastroenteritis, cholecystitis, pancreatitis, hepatitis, peptic ulcer, pyelonephritis; after midpregnancy, **severe preeclampsia and fatty liver**. Endoscopy changed management in none of 49 women.
- **Thyroxine**: raised free T4 is usually a **surrogate for high hCG**, not true thyrotoxicosis; normalizes quickly with hydration and antiemetics.
- **Readmission 25–35%** (more than 1/3 at Parkland); not predictable from patient characteristics. ↑ Anxiety, depression; relapse in 1/4 of those with psychosocial factors; some develop PTSD.
- **Nutrition for recalcitrant cases**:
 - Early enteral tube feeding: **no advantage**.
 - **Nasojejunal feeding** used for up to 41 days (107 women); sonography can confirm tube position. PEG-J reported.
 - Very few need parenteral nutrition (though 20% needed CPN in one study of 599 women).

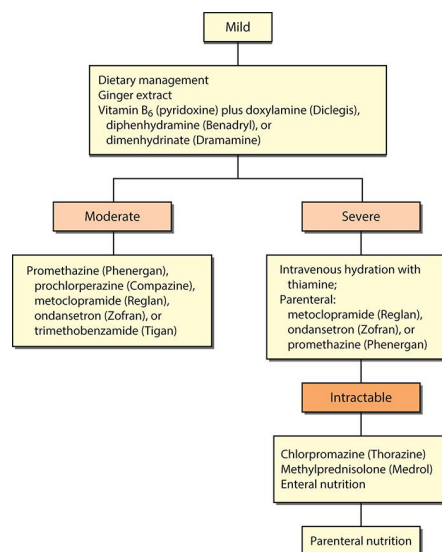


Figure 57-1. Stepwise NVP algorithm: diet/ginger/B6 + doxylamine → oral antiemetics (moderate) → IV fluids with thiamine + parenteral antiemetics (severe) → chlorpromazine, methylprednisolone, enteral feeds (intractable) → parenteral nutrition.

Table 57-4. Medications for Gastrointestinal Disorders in Pregnancy

Options for nausea and vomiting

Class / Medication (Brand)	Usual Dosing	Route(s)
Antihistamine		
Doxylamine + pyridoxine (Diclegis) ^a	At bedtime; up to 4 times daily	PO
Phenothiazines		
Promethazine (Phenergan) ^c	12.5–25 mg	IM, IV, PO, PR
Prochlorperazine (Compazine) ^c	5–10 (25 PR) mg	IM, IV, PO, PR
Chlorpromazine (Thorazine) ^c	25–50 mg	IM, PO
Serotonin antagonist		
Ondansetron (Zofran) ^b	8 mg	IV, PO
Benzamides		
Metoclopramide (Reglan) ^b	5–15 mg	IM, IV, PO
Trimethobenzamide (Tigan)	300 mg	PO

Oral options for gastroesophageal reflux disease (GERD)

Class / Medication (Brand)	Usual Dosing
Proton-pump inhibitors	
Pantoprazole (Protonix) ^b	40 mg daily for up to 8 wks
Lansoprazole (Prevacid) ^b	15 mg daily for up to 8 wks
Omeprazole (Prilosec, Zegerid) ^c	20 mg daily for 4–8 wks
Dexlansoprazole (Dexilant) ^c	30 mg daily for up to 4 wks
H2-receptor antagonists	
Cimetidine (Tagamet) ^b	400 mg 4 times daily for up to 12 wks, or 800 mg twice daily for up to 12 wks
Nizatidine (Axid) ^b	150 mg twice daily
Famotidine (Pepcid) ^b	20 mg twice daily up to 6 wks

^aFood and Drug Administration category A. ^bFDA category B. ^cFDA category C. IM = intramuscularly; IV = intravenously; PO = orally; PR = rectally.

3. Gastroesophageal Reflux Disease

- Symptomatic reflux in up to **15%** of nonpregnant people. Sequelae: esophagitis, stricture, Barrett esophagus, adenocarcinoma.
- **Heartburn (pyrosis) prevalence: 26% → 36% → 51%** in the 1st / 2nd / 3rd trimesters. Caused by relaxation of the lower esophageal sphincter.
- **Diagnosis is clinical** (symptoms alone).
- **Stepwise treatment:**
 1. Lifestyle: stop tobacco and alcohol, small meals, **elevate head of bed**, limit postprandial recumbency, avoid triggers (fatty and tomato-based foods, coffee)
 2. **Oral antacids = first line**
 3. Severe/persistent → **proton-pump inhibitor**; H2-receptor antagonist and sucralfate can be added. Both classes are generally safe.
 4. No relief → consider **endoscopy**
- PPIs early in pregnancy may ↑ preeclampsia risk (preliminary data).
- **Sucralfate:** aluminum salt of sulfated sucrose, inhibits pepsin; **~10% absorbed**, safe. **1 g PO 1 h before each meal and at bedtime for up to 8 wk**; no antacids within ½ h of a dose.
- **Misoprostol is contraindicated** (stimulates labor).
- Fundoplication is not done in pregnancy; after prior laparoscopic Nissen, only **20%** had reflux in pregnancy.

4. Hiatal, Diaphragmatic Hernia and Achalasia

Hiatal hernia

- On late-pregnancy radiographs: **20% of multiparas vs 5% of nulliparas**; persisted in 3 of 10 at 1–18 months postpartum.
- Link with reflux unclear (the LES can work even when intrathoracic).
- Can cause vomiting, epigastric pain, bleeding from ulceration; rarely gastric outlet obstruction (paraesophageal hernia) or surgery needed.

Diaphragmatic hernia

- Through the **foramen of Bochdalek or Morgagni**; rare.
- **Acute obstruction** → **maternal mortality 45%** (18 cases).
- Spontaneous diaphragmatic rupture from ↑ intraabdominal pressure at delivery reported. Surgery individualized by gestational age.

Achalasia

- LES **fails to relax** + nonperistaltic contractions, from inflammatory destruction of the **myenteric (Auerbach) plexus**; postganglionic cholinergic neurons spared → unopposed sphincter stimulation.
- Symptoms: **dysphagia, chest pain, regurgitation**.
- **Barium swallow: "bird beak" or "ace of spades"** distal narrowing. Endoscopy excludes gastric carcinoma; **manometry confirms**.
- Pregnancy: physiological LES relaxation shouldn't occur; half worsened in one series, but most don't; reflux uncommon. One death at 24 wk (perforated **14-cm megaesophagus**).
- **Management**: soft diet + anticholinergics → nitrates, calcium-channel blockers, **botulinum toxin A** → **balloon dilation** (85% respond; risk of perforation) → **myotomy**.

5. Peptic Ulcer Disease

- Lifetime prevalence of acid peptic disease in women **10%**. Causes: **chronic *H pylori* gastritis or NSAIDs** — neither common in pregnancy.
- **Natural gastroprotection in pregnancy**: ↓ gastric acid, ↓ motility, markedly ↑ mucus. Complications are less frequent than in nonpregnant controls; perforation rare. May be underdiagnosed (treated as reflux).
- Historical: **remission in ~90%** during pregnancy, but symptoms recurred in >½ by 3 months postpartum and almost all by 2 years.
- **Management**: eradicate *H pylori*, prevent NSAID injury. First line = **PPI or H2 blocker** (± sucralfate coating).
- ***H pylori* testing** (with active ulcer): urea breath test, serology, fecal test, or endoscopic biopsy.
- **Pregnancy-safe 14-day eradication (no tetracycline)**: clarithromycin 500 mg bid + either amoxicillin 1000 mg bid or metronidazole 500 mg tid, + a PPI.

Upper GI bleeding

- With persistent vomiting, most bleeds are **Mallory-Weiss tears** (linear mucosal tears at the GE junction); occasionally a peptic ulcer.
- Conservative: **IV PPI + topical antacids**; transfuse as needed; **endoscopy if bleeding persists**.
- Sustained retching → **Boerhaave syndrome** (esophageal rupture) — less common, more serious.

6. Small Bowel and Colon Disorders

Motility and constipation

- **Small bowel transit slows: 99 / 125 / 137 min** by trimester vs **75 min** nonpregnant.
- Colonic relaxation + ↑ water and Na⁺ absorption → **constipation in ~25%**.
- Treatment: fluids **>8 glasses/d**, fiber **20–35 g/d** → bulk agents (methylcellulose) → stimulant (bisacodyl) if severe.
- **Megacolon from impacted stool** seen almost only in women who chronically abused stimulant laxatives.

Acute diarrhea

- Monthly prevalence **3–7%** in adults.
- **Classification: acute <2 wk; persistent 2–4 wk; chronic >4 wk.** Most acute cases are infectious; 1/3 foodborne.
- **Indications for evaluation:**
 - Profuse watery diarrhea with dehydration
 - Grossly bloody stools
 - **Fever >38°C**
 - **Duration >48 h** without improvement
 - Recent antimicrobial use
 - Immunocompromise
- **Treatment:**
 - **Mainstay: IV hydration** (normal saline or Ringer lactate) + **K⁺**, to restore volume and uteroplacental perfusion. Monitor vitals and urine output for sepsis.
 - **Loperamide** (moderate, afebrile, nonbloody, no recent travel): **4 mg, then 2 mg after each watery stool; max 8 mg/d; ≤48 h.**
 - **Bismuth subsalicylate:** 30 mL (525 mg) or 2 tablets (263 mg each) every 30–60 min, **max 8 doses/24 h;** harmless black stools.
 - Empirical antibiotics for moderate–severe illness: **ciprofloxacin 500 mg bid × 3 days** (reasonable to wait for stool tests).
 - Usually no treatment needed: *E coli*, staphylococci, *B cereus*, norovirus.
 - **Specific:** *Salmonella* → ciprofloxacin or TMP-SMX; *Campylobacter* → azithromycin; *C difficile* → oral vancomycin or fidaxomicin; *Giardia*, *E histolytica* → metronidazole.

Table 57-5. Etiology, Clinical Features, and Treatment of Common Acute Diarrheal Syndromes

Agents	Incubation	Emesis	Pain	Fever	Diarrhea	Treatment
Toxin producers	1–72 hr	3–4+	1–2+	0–1+	3–4+, watery	
1. <i>Staphylococcus</i> spp.						None
2. <i>C perfringens</i>						None
3. <i>E coli</i> (enterotoxin)						Ciprofloxacin
4. <i>B cereus</i>						None
Enteroadherent	1–8 days	0–1+	1–3+	0–2+	1–2+, watery, mushy	
1. <i>E coli</i>						Ciprofloxacin
2. <i>Giardia</i> spp.						Tinidazole
3. Helminths						As detected
Cytotoxin producers	1–3 days	0–1+	3–4+	1–2+	1–3+, watery, then bloody	
1. <i>C difficile</i>						Vancomycin
2. <i>E coli</i> (hemorrhagic)						None
Inflammatory — Minimal	1–3 days	1–3+	2–3+	3–4+	1–3+, watery	
1. Rotavirus						None
2. Norovirus						None
Inflammatory — Variable	12 hr–11 days	0–3+	2–4+	3–4+	1–4+ watery or bloody	
3. <i>Salmonella</i> spp.						Ciprofloxacin
4. <i>Campylobacter</i> spp.						Azithromycin
5. <i>Vibrio</i> spp.						Doxycycline
Inflammatory — Severe	12 hr–8 days	0–1+	3–4+	3–4+	1–2+, bloody	
6. <i>Shigella</i> spp.						Ciprofloxacin
7. <i>E coli</i>						Ciprofloxacin
8. <i>Entamoeba histolytica</i>						Metronidazole

B cereus = *Bacillus cereus*; *C difficile* = *Clostridioides difficile*; *C perfringens* = *Clostridium perfringens*; *E coli* = *Escherichia coli*; spp. = species. Modified from Camilleri, 2018; DuPont, 2014; McDonald, 2018.

Memory hook: "Toxins vomit, cytotoxins hurt, inflammatory ones burn (fever)." Toxin producers = emesis 3–4+ with short incubation (hours); cytotoxins = pain 3–4+ and bloody diarrhea; inflammatory = fever 3–4+.

***Clostridioides difficile* infection**

- Anaerobic gram-positive bacillus, fecal–oral. **3.7%** of normal gravidas colonized. **Most frequent nosocomial infection in the US**; incidence in pregnancy has **doubled** in a decade.
- **Biggest risk factor = antibiotics** — highest with **aminopenicillins, clindamycin, cephalosporins, fluoroquinolones**. Others: IBD, immunosuppression, age, GI surgery. **10% community-acquired**.
- **Severe colitis mortality 5%.**
- **Diagnosis:** stool **NAAT** or enzyme-immunoassay algorithm (a unique *C difficile* enzyme, or **toxins A and B**); endoscopy less often (pseudomembranes). **Test only with new diarrhea and ≥3 unformed stools** of unexplained cause; **no test of cure**.
- **Prevention: soap-and-water hand washing** + isolation.
- **Treatment:** stop the offending antibiotic + **oral vancomycin 125 mg qid × 10 days** or **fidaxomicin 200 mg bid**.
- **Recurrence 20%**; fecal microbial transplantation may become standard for recurrence.

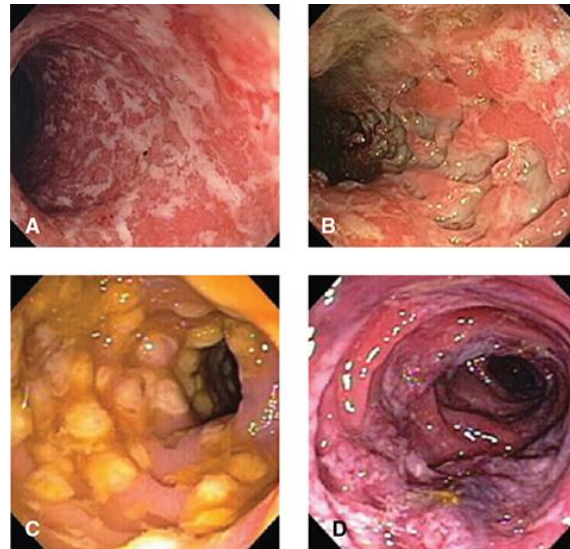


Figure 57-2. Colitis at endoscopy: (A) ulcerative colitis—diffuse ulcers and exudate; (B) Crohn—deep ulcers; (C) pseudomembranous (*C difficile*)—yellow adherent membranes; (D) ischemic colitis.

7. Inflammatory Bowel Disease

Table 57-6. Some Shared and Differentiating Characteristics of Inflammatory Bowel Disease

Shared characteristics

Feature	Both
Hereditary	More than 100 disease-associated genetic loci —a third shared; Jewish predominance; familial in 5–10% of cases; Turner syndrome; immune dysregulation
Other	Chronic and intermittent with exacerbations and remissions; extraintestinal manifestations: arthritis, erythema nodosum, uveitis

Differentiating characteristics

Feature	Ulcerative Colitis	Crohn Disease
Major symptoms	Diarrhea, tenesmus, rectal bleeding , cramping pain; chronic, intermittent; anemia	<i>Fibrostenotic</i> —recurrent RLQ colicky pain; fever. <i>Fistulizing</i> —cutaneous, bladder, interenteric
Bowel involvement	Mucosa and submucosa of large bowel ; usually begins at rectum (40% proctitis only); continuous disease	Deep layers small and large bowel; commonly transmural ; discontinuous involvement; strictures and fistulas
Endoscopy	Granular and friable erythematous mucosa; bleeding; rectal involvement	Patchy; segmental colitis; rectum spared ; perianal involvement
Serum antibodies	Antineutrophil cytoplasmic (pANCA) ~70%	Anti-<i>S cerevisiae</i> ~50%
Complications	Toxic megacolon; strictures; arthritis; cancer (3–5%)	Fistulas; arthritis; toxic megacolon; small bowel obstruction
Management	Medical; proctocolectomy curative	Medical; segmental and fistula resection

RLQ = right lower quadrant; *S cerevisiae* = *Saccharomyces cerevisiae*. From Friedman, 2018; Lichtenstein, 2009.

Ulcerative colitis

- **Mucosal** inflammation of the colon, **starts at the rectum** and extends proximally. **~40%** rectum/rectosigmoid only; **20% pancolitis**.
- **Prior appendectomy protects** against UC.

- Symptoms: diarrhea, rectal bleeding, tenesmus, cramps; exacerbations and remissions.
- Dangerous complications: **toxic megacolon, catastrophic hemorrhage** → colectomy. Extraintestinal: arthritis, uveitis, erythema nodosum.
- **Colon cancer risk approaches 1%/year.**

Crohn disease

- Regional enteritis / granulomatous colitis. **Transmural**, anywhere **mouth to anus, segmental**.
- Distribution: **30% small bowel only, 25% colon only, 40% both** (terminal ileum + colon).
- **Perianal fistulas/abscesses in 1/3** of those with colonic disease.
- Symptoms: RLQ cramping pain, diarrhea, weight loss, low-grade fever, obstruction.
- **Cannot be cured medically or surgically.** ~1/3 need surgery in the first year, then **5%/year**.

IBD and fertility

- **Subfertility linked to active disease**; normal fertility likely when quiescent.
- **Sulfasalazine** → **reversible sperm abnormalities** (male partners).
- Laparoscopic resection → better fertility. After colectomy **up to half remain infertile**. Ileal pouch–anal anastomosis only modestly affects fertility.

IBD and pregnancy (general rules)

- Prevalence **1 in 320 births** (Australia).
- **Pregnancy does NOT increase flares** (rate even lower than preconception). But **activity at conception predicts activity in pregnancy**, and flares can be severe.
- **Continue usual treatment**; investigate and operate if indicated.
- **Fecal calprotectin** is valid for detecting flares in pregnancy.
- Adverse outcomes are concentrated in women with **severe/active disease**; with quiescent disease outcomes are similar to controls (ECCO-EpiCom, 332 women). **Active disease** → **↑ preterm birth**. Perinatal mortality not appreciably ↑.
- **VTE risk ↑** even when asymptomatic (UC: **doubled**), but thromboprophylaxis is **not routine**.

Ulcerative colitis in pregnancy

Disease state at conception	Course in pregnancy
Quiescent	Worsens in ~1/3
Active	~45% worsen, 25% unchanged, 25% improve

- **Osteoporosis** in up to 1/3 → **vitamin D 800 IU/d + calcium 1200 mg/d**.
- **Folic acid 2–4 mg/d** preconception and 1st trimester (counteracts **sulfasalazine's antifolate** action).
- Stress may trigger flares → reassurance.
- **Drug therapy mirrors nonpregnant care:**
 - **5-ASA (mesalamine)**: sulfasalazine (prototype; inhibits prostaglandin synthase), olsalazine, balsalazide, delayed-release forms.
 - **Glucocorticoids** (oral, parenteral, enema) for moderate–severe disease not responding to 5-ASA — **high remission rate, but not for maintenance**.
 - **Immunomodulators** for recalcitrant disease: azathioprine, 6-MP, cyclosporine, tacrolimus — relatively safe.
 - **⚠ Methotrexate is teratogenic → contraindicated.**
 - **Biologics (anti-TNF-α etc.)** now often used first for moderate–severe disease. **Continue through pregnancy until several weeks before delivery** (Table 57-7). Stopping may prompt relapse.
 - **Neonate:** possible immunosuppression → **delay live-attenuated vaccines ≥6 months**.

- **Surgery:** colectomy/ostomy for fulminant colitis is lifesaving and has been done in every trimester, generally with good outcomes (42 cases).
- **Ileal pouch:** ↑ stool frequency, incontinence and **pouchitis** (bacterial overgrowth, stasis) → **ciprofloxacin or metronidazole**. Rarely pouch perforation from adhesions.
- **Delivery:**
 - Uncomplicated colitis → **vaginal delivery**; but cesarean rate is ↑.
 - **After ileal pouch–anal anastomosis: controversial.** Hahnloser: similar function → cesarean only for obstetric indications. Foulon review: recommends cesarean, though uncomplicated vaginal delivery is safe.
- Quiescent UC: abortion, preterm and stillbirth rates remarkably low (2398 pregnancies).

Table 57-7. Biologics Used to Treat Inflammatory Bowel Disease in Pregnancy

Drug (Brand Name)	Recommendations
Adalimumab (Humira)	Last dose 2–3 wks before EDC
Certolizumab pegol (Cimzia)	Continue throughout pregnancy
Infliximab (Remicade)	Last dose 6–10 wks before EDC
Natalizumab (Tysabri)	Last dose 4–6 wks before EDC
Ustekinumab (Stelara)	Last dose 6–10 wks before EDC
Vedolizumab (Entyvio)	Last dose 6–10 wks before EDC

EDC = estimated date of confinement.

Pearl: certolizumab pegol lacks an Fc region, so it crosses the placenta minimally — which is why it alone can be continued throughout pregnancy.

Crohn disease in pregnancy

Disease state at conception	Course in pregnancy
Inactive	~¼ relapse
Active	⅔ remain active or worsen

- Calcium, vitamin D and folic acid as for UC.
- Maintenance: no universally effective regimen; 5-ASA (newer forms better tolerated than sulfasalazine).
- **Glucocorticoids** → **60–70% remission**; prednisone less effective for small-bowel disease.
- Immunomodulators (azathioprine, 6-MP, cyclosporine, tacrolimus) relatively safe; **methotrexate contraindicated**; **TNF-α inhibitors** often used first.
- **Surgery** for fistulas, strictures, obstruction, abscess, intractable disease (small-bowel disease more likely); abdominal surgery needed in **5%** of pregnancies.
- **Delivery: vaginal unless perianal fistula or active perianal disease.**
- Outcomes worse than with UC, related to activity: **~2× preterm** (Danish), **2–3× preterm, LBW, FGR, cesarean** (149 women). ECCO-EpiCom: similar to controls.

Ostomy and pregnancy

- **Stomal dysfunction common** but usually responds to conservative care (82 pregnancies).
- Surgery needed in 3 of 6 with bowel obstruction and 4 with **ileostomy prolapse** — **~10% overall**.
- Adhesions usually cause obstruction, but the enlarging uterus may too. Pregnancy does **not** worsen long-term ostomy function.

8. Intestinal Obstruction

- **Incidence not increased** in pregnancy, but harder to diagnose: **~1 in 17,000** deliveries.
- Among acute abdomens in pregnancy: **appendicitis 30% > adhesive small-bowel obstruction 15%**.
- **Causes:**
 - **Adhesions ~50%** (prior pelvic surgery, including cesarean)
 - **Volvulus ~25%** (sigmoid, cecum, small bowel) — late pregnancy or early puerperium
 - Others: after **Roux-en-Y gastric bypass** (internal hernia/intussusception), intussusception, ventral hernia. Open (vs laparoscopic) colorectal surgery → 3× obstruction.
- **High-risk times** (uterus pressing on adhesions):
 1. **Midpregnancy** — uterus becomes an abdominal organ
 2. **3rd trimester** — fetal head descends
 3. **Immediately postpartum** — uterus shrinks acutely
- **Presentation:** abdominal pain (continuous or colicky) **98%**; nausea/vomiting **80%**; tenderness **70%**; abnormal bowel sounds only **55%**.
- **Imaging:** plain films after soluble contrast show obstruction in **90%**, but are less accurate for small bowel → **CT and MR can be diagnostic. Colonoscopy can diagnose and treat colonic volvulus.**
- **Mortality is excessive** from delayed diagnosis, reluctance to operate, and emergency surgery: **maternal 6%, fetal 26%** (66 pregnancies); deaths in late pregnancy from perforated sigmoid/cecal volvulus.

Colonic pseudo-obstruction (Ogilvie syndrome)

- **Adynamic colonic ileus** → massive distention with **cecal and right-colon dilation**.
- **~10%** of all cases are pregnancy-associated; **1 in 1500 births**.
- Usually **postpartum — 90% after cesarean**; also antepartum.
- **Perforation risk high if cecal diameter >12 cm.**
- **Treatment: IV neostigmine 2 mg with cardiac monitoring** → prompt decompression. Otherwise colonoscopic decompression; laparotomy for perforation.

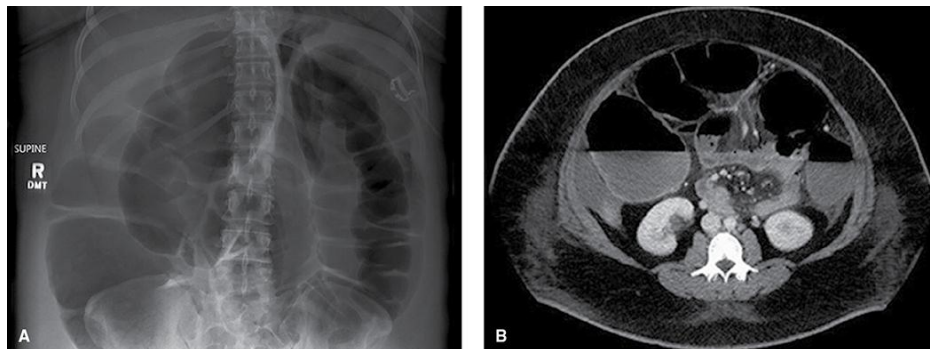


Figure 57-3. Ogilvie syndrome after cesarean: (A) plain film and (B) CT show massive gaseous colonic dilation without a mechanical obstruction.

9. Appendicitis

Epidemiology

- Lifetime incidence **7–10%**. Confirmed appendicitis in pregnancy **1 in 1000 to 1 in 5500 births** (only 14 of 171 evaluated women had it).
- **Most common reason for lifesaving GI surgery in pregnancy.**

Why diagnosis is harder

- Nausea and vomiting are normal in pregnancy.
- **The appendix moves upward and outward** from the RLQ as the uterus grows.
- Physiological leukocytosis — but **WBC >18,000/ μ L makes appendicitis 10x more likely**; a neutrophilic shift also helps.
- Often confused with **cholecystitis, labor, pyelonephritis, renal colic, placental abruption, leiomyoma degeneration**.
- **Omental containment fails** as the appendix is pushed up → **perforation $\uparrow$ with gestational age: ~8%, 12%, 20%** by trimester. Morbidity and mortality rise with gestational age.

Diagnosis

- **Persistent abdominal pain and tenderness** are the most reproducible findings; **RLQ pain most frequent**, but pain migrates upward.
- **Sonography first** (also excludes obstetric causes); graded compression is difficult.
- **CT** is more sensitive and accurate than sonography; views can limit fetal dose.
- **MR imaging is preferred when available** — high accuracy and gives alternative diagnoses. Meta-analysis: **PPV 96%, NPV 99.5%** (but one study: sensitivity only 18%).

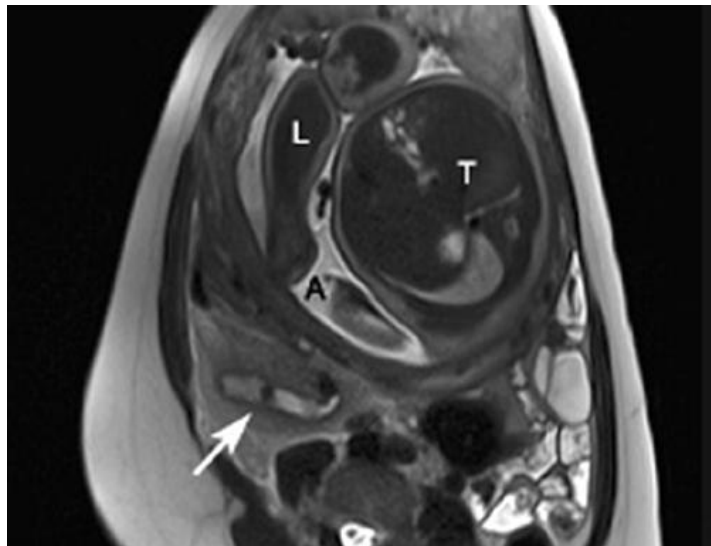


Figure 57-4. MR (T2) of midtrimester acute appendicitis: bright inflamed appendix with a dark fecalith (arrow) below the gravid uterus (T = fetal trunk, L = leg, A = amnionic fluid).

Management

- **Suspected appendicitis → prompt surgical exploration.** A negative appendectomy is preferable to delayed surgery and generalized peritonitis.
 - Historically the diagnosis was confirmed in only **60–70%**; CT/MR have improved this. **Diagnostic accuracy is inversely proportional to gestational age.**
- **Laparoscopy is almost always used in the 1st–2nd trimesters**, and in many centers also in the 3rd (SAGES supports). Evidence does not clearly favor laparoscopic vs open for fetal outcome.
- **Preoperative IV antibiotics:** 2nd-generation cephalosporin or 3rd-generation penicillin. Stop after surgery unless **gangrene, perforation or phlegmon**.
- Without generalized peritonitis, **prognosis is excellent**.
- **Cesarean at the time of appendectomy is seldom indicated.**
- Uterine contractions are common, but **tocolytics are NOT recommended** — they $\uparrow$ risk of **pulmonary permeability edema** from sepsis.

- **Antibiotics alone (nonoperative):** in pregnancy linked to ↑ septic shock, peritonitis and VTE (6% treated medically); **25% failure** in 20 women. **Discouraged** in pregnancy; if used, exclude obstructive appendicitis and keep a low threshold to operate.

Pregnancy outcomes

- ↑ **Preterm labor**, especially with **peritonitis**. Spontaneous labor after 23 wk is commoner after appendectomy than after other surgery.
- **Fetal loss 22%** if surgery after 23 wk; **20%** in the California database.
- LBW and preterm birth ↑ **1.5–2-fold** (Taiwan); preterm birth ~2× (Australia).

Quick-Recall: Condition → Key Diagnostic Clue → Key Management

Condition	Key clue / fact	Management pearl
Hyperemesis gravidarum	Weight loss, ketosis, hypokalemic alkalosis ; diagnosis of exclusion	Doxylamine–pyridoxine → ondansetron (>8 wk) → IV fluids + thiamine 100 mg → steroids only if refractory
Wernicke encephalopathy	Ocular signs + confusion + ataxia (~60%); MR findings	Thiamine before/with IV fluids if vomiting >3 wk
GERD	Heartburn 26 → 36 → 51% by trimester	Lifestyle → antacids → PPI/H2RA/sucralfate; no misoprostol
Achalasia	Bird-beak on barium swallow; manometry confirms	Diet/anticholinergics → nitrates/CCB/botox → balloon dilation → myotomy
Peptic ulcer	Rare in pregnancy; ~90% remit	PPI/H2RA; H pylori: clarithromycin + amoxicillin or metronidazole + PPI × 14 d
Mallory-Weiss vs Boerhaave	Linear GE-junction tear vs full-thickness rupture	IV PPI, antacids, endoscopy if persistent vs surgical emergency
<i>C difficile</i>	New diarrhea + ≥3 unformed stools, recent antibiotics	Stop antibiotic; oral vancomycin 125 mg qid × 10 d or fidaxomicin
Ulcerative colitis	Continuous from rectum, mucosal, pANCA 70%	5-ASA, steroids, thiopurines, biologics; no methotrexate ; vaginal birth usual
Crohn disease	Transmural, skip lesions, rectum spared, ASCA 50%	Same drugs; cesarean if active perianal disease/fistula
Bowel obstruction	Pain 98%, vomiting 80%; mid-pregnancy, descent, postpartum	CT/MR; surgery without delay; volvulus → colonoscopy
Ogilvie syndrome	Cecal dilation, 90% after cesarean	IV neostigmine 2 mg with monitoring; perforation risk if cecum >12 cm
Appendicitis	Persistent pain; appendix displaced up; WBC >18,000	MR preferred ; prompt (laparoscopic) appendectomy; no tocolytics

Key Numbers to Memorize

- Hyperemesis admissions **0.5–2.1%**; recurrence of admission **up to 25%**; readmission **25–35%**
- Diclegis = **doxylamine 10 mg + pyridoxine 10 mg**, 2 at bedtime, max 4/d; **thiamine 100 mg/L** if vomiting >3 wk; ondansetron **after 8 wk**
- CPN **24–40 kcal/kg/d**; catheter sepsis **25%**; PICC complications **56%**
- Heartburn **26 / 36 / 51%**; sucralfate **1 g** qid, ~10% absorbed
- Hiatal hernia **20% multiparas vs 5% nulliparas**; obstructed diaphragmatic hernia mortality **45%**
- Diarrhea: acute <2 wk, persistent 2–4 wk, chronic >4 wk; loperamide max **8 mg/d, 48 h**
- *C difficile*: vancomycin **125 mg qid × 10 d**; recurrence **20%**; severe colitis mortality **5%**

- UC: **40%** proctitis only, **20%** pancolitis; cancer **~1%/yr**; pANCA **70%**. Crohn: **30/25/40%** small bowel/colon/both; ASCA **50%**
- Quiescent UC worsens **1/3**; active UC worsens **45%**; inactive Crohn relapses **1/4**; active Crohn stays active/worse **2/3**
- IBD supplements: vitamin D **800 IU**, calcium **1200 mg**, folic acid **2–4 mg**; live vaccines in exposed neonates delayed **≥6 months**
- Obstruction: **1 in 17,000**; adhesions **50%**, volvulus **25%**; maternal/fetal mortality **6% / 26%**
- Ogilvie: **1 in 1500 births**, **90%** post-cesarean, cecum **>12 cm**, neostigmine **2 mg IV**
- Appendicitis: **1 in 1000–5500**; perforation **8 / 12 / 20%** by trimester; WBC **>18,000** (10×); MR PPV/NPV **96 / 99.5%**; fetal loss **~20–22%** after 23 wk